

Proposed Workflow: Wolffia SMART-seq Cell Atlas Project

Project Goal

Use SMART-seq single-cell RNA-seq data from *Wolffia australiana* to identify major cell populations, propose candidate cell identities, and explore developmental relationships between cell states.

Core Biological Questions

1. How many transcriptionally distinct cell populations can be detected in Wolffia?
2. Does Wolffia show a reduced cellular landscape consistent with its simplified morphology?
3. Are there transitional or previously uncharacterized cell states involved in growth and development?

Suggested Computational Workflow

1. Raw data processing

- Start from per-cell SMART-seq FASTQ files.
- Run read-level QC.
- Align reads to the Wolffia reference genome.
- Generate a gene-by-cell count matrix.

2. Cell-level quality control

- Evaluate reads per cell, detected genes, mapping rate, and organellar transcript fractions.
- Flag low-quality cells and possible technical outliers.
- Review potential batch or plate effects.

3. Exploratory analysis

- Normalize expression values.
- Identify highly variable genes.
- Perform PCA and UMAP for visualization.

4. Clustering and markers

- Cluster cells across a reasonable range of parameters.
- Identify marker genes for each cluster.
- Assess whether clusters are stable and biologically interpretable.

5. Annotation

- Compare marker genes with known plant developmental markers and homologs.
- Assign conservative candidate labels where possible.
- Keep uncertain populations labeled as provisional or unknown.

6. Trajectory analysis

- Use graph-based or pseudotime methods to examine developmental relationships.
- Identify possible transition states or branch points.

7. Project outputs

- Reproducible analysis pipeline.
- QC summary and clustering figures.
- Marker gene tables.
- Preliminary cell atlas with candidate annotations.
- Hypotheses for future biological validation.

Feasibility Notes

- This project is strongest if new SMART-seq data will be generated in the lab.
- If the experimental data are not available yet, public single-cell plant datasets could still be used to practice pipeline development and analysis methods.
- Annotation may be challenging if *Wolffia* reference annotations and marker knowledge are limited, so early conclusions should remain conservative.